

An Exact Dimension-Four Reduction of a Twisted-Convolution Identity to a Shintani Special Value

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Abstract

We study the rank condition that turns canonical Shintani–Faddeev real-multiplication values into ghost projectors for Weyl–Heisenberg symmetric informationally complete measurements. We give exact Zak-rank, exterior-square, and parity-moment formulations for the canonical family. In dimension four, all 36 minors of one explicitly defined normalized ghost matrix are divisible by the common factor $x^2 - \sqrt{3 + \sqrt{5}}x + 1$. Using the Shintani–Kopp Kronecker-limit formula requires additional ray-class and multiplier identifications which we state explicitly but do not prove here. Independently, the analytic class-number formula for $\mathbb{Q}(\sqrt{5}, \sqrt{\phi})$ gives the target value $\exp L'(0, \chi) = \phi + \sqrt{\phi}$. Consequently we prove, unconditionally, that the displayed special-value identity implies rank one of the explicit ghost. We do not claim the full two-shift Twisted Convolution Conjecture.

1 Canonical setup

For $d \geq 4$, put

$$Q_d = \langle 1, 1 - d, 1 \rangle, \quad L_d = \begin{pmatrix} d-1 & -1 \\ 1 & 0 \end{pmatrix}.$$

Then

$$\text{disc}(Q_d) = (d+1)(d-3), \quad L_d^2 + L_d + I = dL_d,$$

so $L_d^3 \equiv I \pmod{d}$. Its positive quadratic fixed point is

$$\beta_d = \frac{d-1 + \sqrt{(d+1)(d-3)}}{2}.$$

For $D_{p,q} = \tau^{pq} X^p Z^q$, $\tau = -e^{\pi i/d}$, write

$$P = \frac{1}{d} \sum_{\mathbf{p}} a_{\mathbf{p}} D_{\mathbf{p}}.$$

The equation $P^2 = P$ is a finite twisted-convolution equation for the coefficients. In even dimensions, the representative-wrap signs of the displacement operators are retained. The principal-ghost and formal TCC constructions motivating this normalization are those of Appleby–Flammia–Kopp [2]; software is not used as their mathematical source.

2 Finite rank certificates

After a scalar shift, shear, and partial Fourier transform, canonical twisted convolution is equivalent to rank one of a $d \times d$ matrix K_d . Thus it is equivalent to the vanishing of all 2-minors.

For any complex matrix K ,

$$\Delta_2(K) = \frac{1}{2} \left((\text{Tr } K^\dagger K)^2 - \text{Tr}((K^\dagger K)^2) \right) = \sum_{\substack{|I|=2 \\ |J|=2}} |\det K_{I,J}|^2.$$

Hence $\Delta_2(K) = 0$ if and only if $\text{rank } K \leq 1$.

The ghost also obeys parity-Hermiticity $G^\dagger = PGP$. Therefore $J = PG$ is Hermitian and the rank-one certificate becomes

$$\text{Tr } J^4 = (\text{Tr } J^2)^2.$$

Reciprocity forces lower trace information but does not force this fourth moment.

3 Dimension four and conventions

Let

$$\phi = \frac{1 + \sqrt{5}}{2}, \quad \beta = \phi^2, \quad \tau = -e^{\pi i/4}.$$

We use the reciprocal of Kopp's convention:

$$S_2^{\text{here}}(z \mid \omega_1, \omega_2) := \text{Sin}_{2, \text{Kopp}}(z; \omega_1, \omega_2)^{-1} = \frac{\Gamma_2(\omega_1 + \omega_2 - z \mid \omega_1, \omega_2)}{\Gamma_2(z \mid \omega_1, \omega_2)}.$$

This definition is decisive: replacing S_2^{here} by Kopp's Sin_2 inverts the special value below.

The normalized signed double-sine table is

$$\begin{pmatrix} \sqrt{5} & -x & 1 & -x^{-1} \\ -x^{-1} & -x^{-1} & -x^{-1} & -x \\ 1 & -x^{-1} & 1 & x \\ -x & -x^{-1} & x & -x \end{pmatrix}.$$

The zero characteristic is exceptional and is replaced by $\sqrt{5}$ before applying the displacement chirp. Every table entry is then divided by $\sqrt{5}$ in forming the normalized Weyl overlaps; in particular the exceptional entry gives $a_0 = 1$.

Define the matrix used in this paper, without reference to software, by

$$K_{r,c}(x) = \frac{1}{4\sqrt{5}} \sum_{\substack{0 \leq a, b < 4 \\ r \equiv c+a \pmod{4}}} T_{a,b}(x) \tau^{ab} i^{bc}, \quad 0 \leq r, c < 4,$$

where $T(x)$ is the displayed table and $\tau = -(1+i)/\sqrt{2}$. Reduction of $r = c + a$ modulo four is performed before evaluating the entry; the factor τ^{ab} retains the even-dimensional displacement sign convention. This formula is the complete definition of the matrix whose minors are certified below.

Proposition 1. *Every one of the 36 two-by-two minors of the dimension-four matrix $K(x)$ lies in the principal ideal*

$$\left(x^2 - \sqrt{3 + \sqrt{5}} x + 1 \right)$$

of $\mathbb{Q}(i, \sqrt{2}, \sqrt{5})[x, x^{-1}]$.

Proof. The exact Laurent expansion has 34 formally nonzero minors and two structurally zero minors. Reduce each power using

$$x^{-1} = t - x, \quad x^2 = tx - 1, \quad t = \sqrt{3 + \sqrt{5}} = \frac{\sqrt{2} + \sqrt{10}}{2}.$$

Every remainder $A + Bx$ is zero. More explicitly, for every row pair I and column pair J , the supplementary certificate records a Laurent polynomial $M_{I,J}$ and a polynomial $Q_{I,J}$ satisfying

$$x^2 M_{I,J}(x) = (x^2 - tx + 1) Q_{I,J}(x)$$

coefficientwise in the basis

$$1, \sqrt{2}, \sqrt{5}, \sqrt{10}, t, \sqrt{2}t, \sqrt{5}t, \sqrt{10}t$$

and separately in real and imaginary parts. Thus the certificate is verified by rational arithmetic only. It contains all 36 minors, including the two zero quotients, rather than a Boolean summary. $\square$

It remains to prove $x + x^{-1} = \sqrt{3 + \sqrt{5}}$. Shift and reflection give

$$x = \sqrt{2} \frac{S_2^{\text{here}}(\beta/4 \mid \beta, 1) S_2^{\text{here}}(1/4 \mid \beta, 1)}{S_2^{\text{here}}((\beta + 1)/4 \mid \beta, 1)}.$$

4 The modulus-four ray field

Let $K = \mathbb{Q}(\sqrt{5})$ and $\mathcal{O}_K = \mathbb{Z}[\phi]$. Since 2 is inert, there are $|(\mathcal{O}_K/4)^\times| = 12$ finite residue units. The images of $\{\pm\phi^n\}$, together with the sign at ∞_2 , have order 12. Since $h_K = 1$, the ray exact sequence therefore gives

$$|\text{Cl}_{(4)\infty_2}(K)| = 2.$$

Set $L = K(\sqrt{\phi})$. Its relative discriminant is (4), and the second real place ramifies because $\phi' < 0$. Hence L is the full ray field of modulus $(4)\infty_2$.

Put $u = \phi + \sqrt{\phi}$, and define the nontrivial $\sigma \in \text{Gal}(L/K)$ by $\sigma(\sqrt{\phi}) = -\sqrt{\phi}$. Then $u^\sigma = \phi - \sqrt{\phi} = u^{-1}$, so

$$N_{L/K}(u) = 1, \quad \text{Tr}_{L/K}(u) = 1 + \sqrt{5}.$$

5 Class number, units, and regulator

The field L has signature $(2, 1)$ and discriminant -400 . Its Minkowski bound is $15/(2\pi) < 2.4$. There is no ideal of norm two, hence $h_L = 1$.

Write $s = \sqrt{\phi}$, so $s^4 - s^2 - 1 = 0$. Exact Buchmann class-group/unit computation in PARI/GP, whose input and transcript are included in the certificate package, returns fundamental units $s^3 - s$ and $-s^2 + s$. Since

$$s^3 - s = s^{-1}, \quad -s^2 + s = -u^{-1},$$

these generate the same rank-two lattice as s and $u = s^2 + s$; the exponent-change matrix has determinant one. Consequently

$$\mathcal{O}_L^\times = \{\pm s^m u^n : m, n \in \mathbb{Z}\}, \quad R_L = \log \phi \log u.$$

The exact computation also returns class number one, ray structures $[2]$ and $[2, 2]$, and the same regulator.

For the nontrivial quadratic character χ of L/K ,

$$L(s, \chi) = \frac{\zeta_L(s)}{\zeta_K(s)}.$$

For a number field F , the analytic class-number formula at zero is

$$\zeta_F(s) = -\frac{h_F R_F}{w_F} s^{r_1+r_2-1} + O(s^{r_1+r_2}).$$

Since $h_K = h_L = 1$, $w_K = w_L = 2$, $R_K = \log \phi$, and the respective orders of vanishing are one and two, division of these leading terms yields

$$L'(0, \chi) = \log u.$$

6 The Kronecker-limit bridge

The two-infinite-place ray group has structure $[2, 2]$, while the one-infinite-place group has structure $[2]$. The quotient fibers therefore have order two, which is consistent with $n = 1$ in Kopp's Theorem 1.1. This order calculation alone does not specialize that theorem. A complete specialization must also exhibit the labeled modulus and embeddings, a ray class A , a characteristic $\mathbf{r}$, a stabilizer $B \in \Gamma_{\mathbf{r}}$, the multiplier $(\psi^{-2} \chi_{\mathbf{r}}^{-1})(B)$, and an equality (including sign) $Z(s, A) = L(s, \chi)$. Finally it must convert $\text{shin}_{\mathbf{r}}^{\mathbf{r}}(\beta)^2$ into the three S_2^{here} factors defining x^2 . Those data are not derived here. We therefore record

$$x^2 = \exp L'(0, \chi) = u \tag{KL}$$

as the exact missing bridge, not as a proved equality.

Theorem 2 (Unconditional finite reduction). *Let $K(x)$ be the explicitly defined matrix above. If*

$$\left(\sqrt{2} \frac{S_2^{\text{here}}(\beta/4 \mid \beta, 1) S_2^{\text{here}}(1/4 \mid \beta, 1)}{S_2^{\text{here}}((\beta+1)/4 \mid \beta, 1)} \right)^2 = \phi + \sqrt{\phi}, \tag{SV}$$

then $K(x)$ has rank exactly one.

Proof. Under (SV), $x^2 = u$. Since $u + u^{-1} = 1 + \sqrt{5}$,

$$(x + x^{-1})^2 = 3 + \sqrt{5}.$$

The defining reciprocal double-sine value is positive, hence $x + x^{-1} = \sqrt{3 + \sqrt{5}}$. The common minor factor vanishes, so all 36 minors vanish by Proposition 1. Thus $\text{rank } K(x) \leq 1$. The normalized zero Weyl coefficient is one, and the Weyl-to-matrix Fourier transform is injective, so $K(x) \neq 0$ and its rank is exactly one. $\square$

Corollary 3 (Conditional ghost statement). *If, in addition, a convention-matched specialization of Kopp's Theorem 1.1 proves (KL) for the displayed S_2^{here} value and the principal-ghost construction identifies $K(x)$ with one of its shifted Zak matrices, that ghost is a rank-one projector.*

7 Scope and review points

Theorem 2 concerns one explicitly defined dimension-four matrix. It does not prove both shifts $\lambda = 0, 1$ required by the formal Twisted Convolution Conjecture; no equivalence between those shifts is assumed. Nor does it prove the Minimalist RM Values Conjecture, the Stark conjecture, or Zauner's conjecture. The remaining analytic task is precisely the convention-matched specialization listed in the preceding section.

References

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- [4] T. Shintani, *On certain ray class invariants of real quadratic fields*, J. Math. Soc. Japan 30 (1978), 139–167.
- [5] S. Flammia et al., *Zauner.jl*, <https://github.com/sflammia/Zauner.jl>.